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Faculty of Computer Science | Chair of Distributed and Networked
Systems

Advanced Computer Networks

Introduction

Winter Term 2025/26

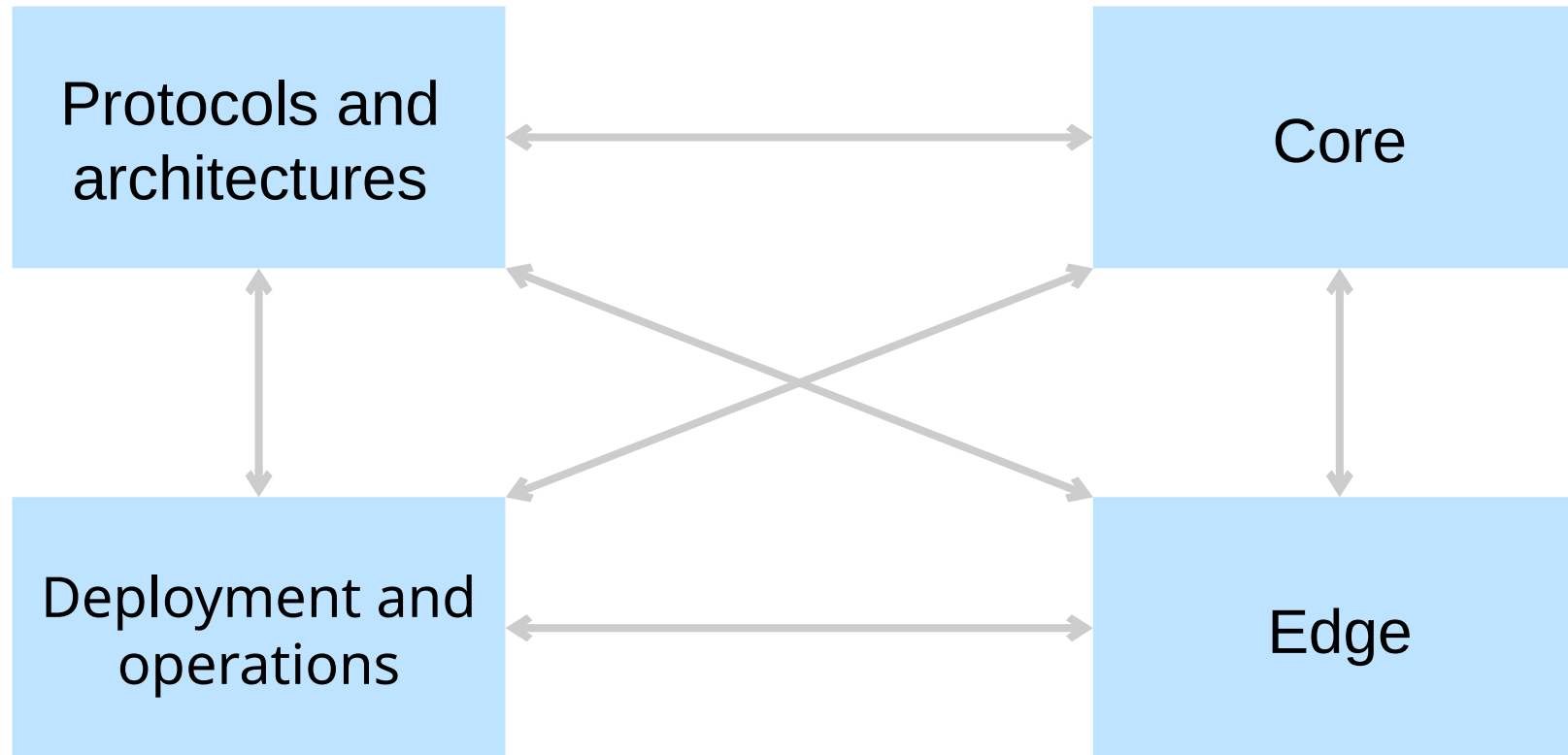
Chair of Distributed and Networked Systems

<https://netd.cs.tu-dresden.de>



Our mission is to advance distributed and networked systems to positively impact the lives of people.

We take a comprehensive view to improve distributed and networked systems.



Communication channel

[matrix]

Mandatory Matrix room: acn-ws25/26

How to get access?

1. **Join the netd-teaching space**
<https://matrix.to/#/#acn-ws2526:tu-dresden.de>
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2. **Join the room asap.**

You can only join as a member of tu-dresden.de.

Time schedule and assignments



Alex



Maynard



Jonas

Lectures

- **Thursday 11:10-12:40, APB/E023**
- **Given by Prof. Wählisch / Feldmann**
- **We will have guest lectures from industry.**

Lab and assignments

- **Wednesday 11:10-12:40, APB/E008**
- **Several assignment sheets**
- **Assignments will include practical work**

Exam

- **TBA** (Don't forget to register in time.)

What is difference compared to the Lecture Computer Networks

Lecture Computer Networks taught you basic concepts

Advanced Computer Networks will focus on practical deployments, considering LANs, Data Center Networks, and the Internet.

After this lecture you should be able

- to have a better systems understanding of networked applications.
- to configure and manage several aspects of computer networks.
- to understand emerging and state-of-the-art network technologies.

**This means that you will do practical work in the exercises.
A Linux environment is required.
You should be able to run docker.**

If you need equipment, contact us.

Syllabus

- **Intro and Recap**
- **Network and transport layer**
- **Remote participation in IETF meeting** [Nov 6]
- **Core Services** (DNS(SEC), Email)
- **System aspects**
- **Inter-domain Routing Security** (BGP, RPKI, BGPsec)
- **How to operate a data center?**
- **Wireshark**
- **Network Virtualization** (Basics, VxLAN, eVPN, OVS, OVN)
- **High-Availability for DC infrastructures** (HA Basics, VRRP, ECMP, BGP for Routing in Large-Scale Data Centers)
- **Network Management** (Basics, SNMP, IPFIX/sFlow, Monitoring/Alerting via Prometheus, Configuration Management)
- **Selected Security Topics** (Basics, VPNs: IPsec, WireGuard, Firewalling (iptables, netfilter, overview eBPF))
- **Traffic Control and High-Performance Data Paths** (tc, basics eBPF, XDP)
- **Solar System Internet** (DTN Basics, BPv7, Contact Graphs, CGR, LTP)